

- Secondary Memory (hard disks, optical memory in DVD-ROM, using which different user programs can be loaded into the Primary memory and subsequently run.
- Input/output Units such as touch screen, modem, etc.
- Input units such as the mouse, keyboard, scanner
- The Operating System
- Networking units such as Ethernet Card, bus drivers, etc.
- User Interfaces and application software which are mostly in secondary memory



The innards of a PC

What is an embedded system

Now have a look at an embedded system. There are three main components of an embedded system.

- It embeds hardware so as to give computer-like functionalities to the device.
- It embeds main application software generally into a flash or ROM and so the application software performs concurrently the number of tasks.
- It embeds a Real Time Operating System, or an RTOS. The RTOS is basically a very, very efficient boss. It enables the execution of concurrent tasks. It also provides a modus operandi by which it allows the processor (who in this analogy is the employee) to run each process (or 'project') as per scheduling and to switch between various process depending on current priority levels. The RTOS, in effect, sets the framework of rules during the execution of application processes so as to enable finishing of a process in good time and with the appropriate priority.

Getting the picture? Embedded systems have also been defined as devices that include a programmable computer but by itself are not intended to serve as general purpose computers, or to elaborate further, the computer is hidden or embedded in the system. Any device that seems to be intelligent and is not a computer can generally be assumed to be an embedded system. The gate which opens automatically when a car is closing in on it? Embedded system. A valve which opens only when the humidity in the room is at a certain level? Embedded system. But then, we hear you wonder, can we have an embedded Operating System ON a PC?